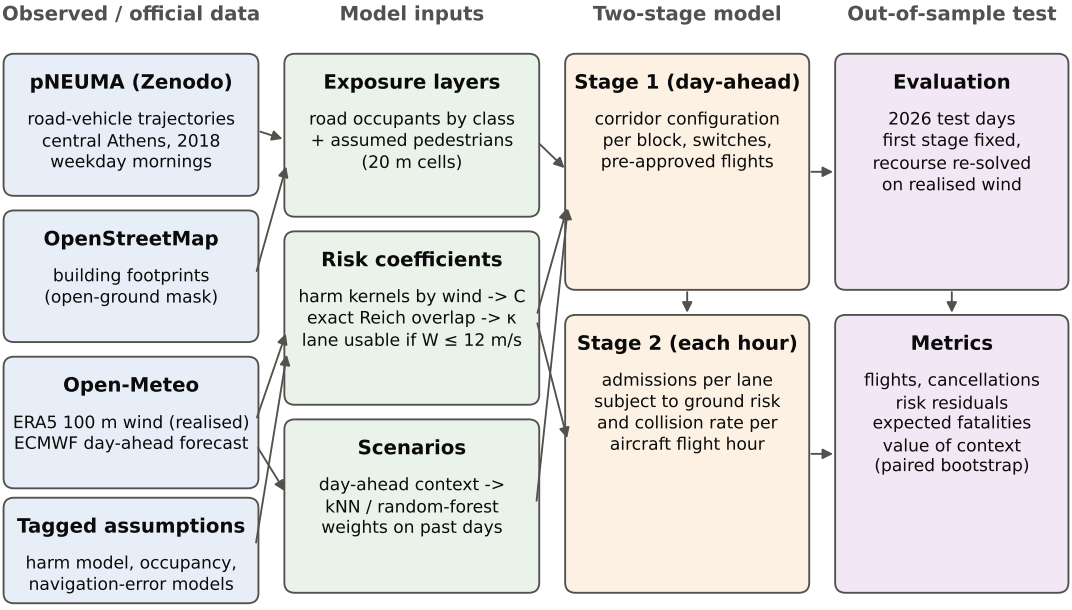
Observed / official data
Model inputs
Two-stage model
Out-of-sample test
pNEUMA (Zenodo)
road-vehicle trajectories
central Athens, 2018
weekday mornings
OpenStreetMap
building footprints
(open-ground mask)
Open-Meteo
ERA5 100 m wind (realised)
ECMWF day-ahead forecast
Tagged assumptions
harm model, occupancy,
navigation-error models
Exposure layers
road occupants by class
+ assumed pedestrians
(20 m cells)
Risk coefficients
harm kernels by wind -> C
exact Reich overlap -> κ
lane usable if W ≤ 12 m/s
Scenarios
day-ahead context ->
kNN / random-forest
weights on past days
Stage 1 (day-ahead)
corridor configuration
per block, switches,
pre-approved flights
Stage 2 (each hour)
admissions per lane
subject to ground risk
and collision rate per
aircraft flight hour
Evaluation
2026 test days
first stage fixed,
recourse re-solved
on realised wind
Metrics
flights, cancellations
risk residuals
expected fatalities
value of context
(paired bootstrap)